Experiment 19

High Level Programming Extensions – Triggers

Aim

To create and execute a MySQL trigger that automatically records employee salary changes in a log table.

Software Required

MySQL Workbench or MySQL Command Line Client.

SQL Program:

Step 1: Create database

```
CREATE DATABASE CompanyDB;
USE CompanyDB;
```

Step 2: Create tables

```
CREATE TABLE Employee (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    salary DECIMAL(10,2)
);

CREATE TABLE Salary_Log (
    log_id INT AUTO_INCREMENT PRIMARY KEY,
    emp_id INT,
    old_salary DECIMAL(10,2),
    new_salary DECIMAL(10,2),
    changed_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

Step 3: Create trigger

```
DELIMITER //
CREATE TRIGGER salary_update
AFTER UPDATE ON Employee
FOR EACH ROW
BEGIN
    INSERT INTO Salary_Log (emp_id, old_salary, new_salary)
    VALUES (OLD.emp_id, OLD.salary, NEW.salary);
END //
DELIMITER ;
```

Step 4: Insert and update sample data

```
INSERT INTO Employee VALUES
(1, 'Ravi', 25000),
(2, 'Priya', 30000);

UPDATE Employee
SET salary = 35000
WHERE emp_id = 1;
```

Queries

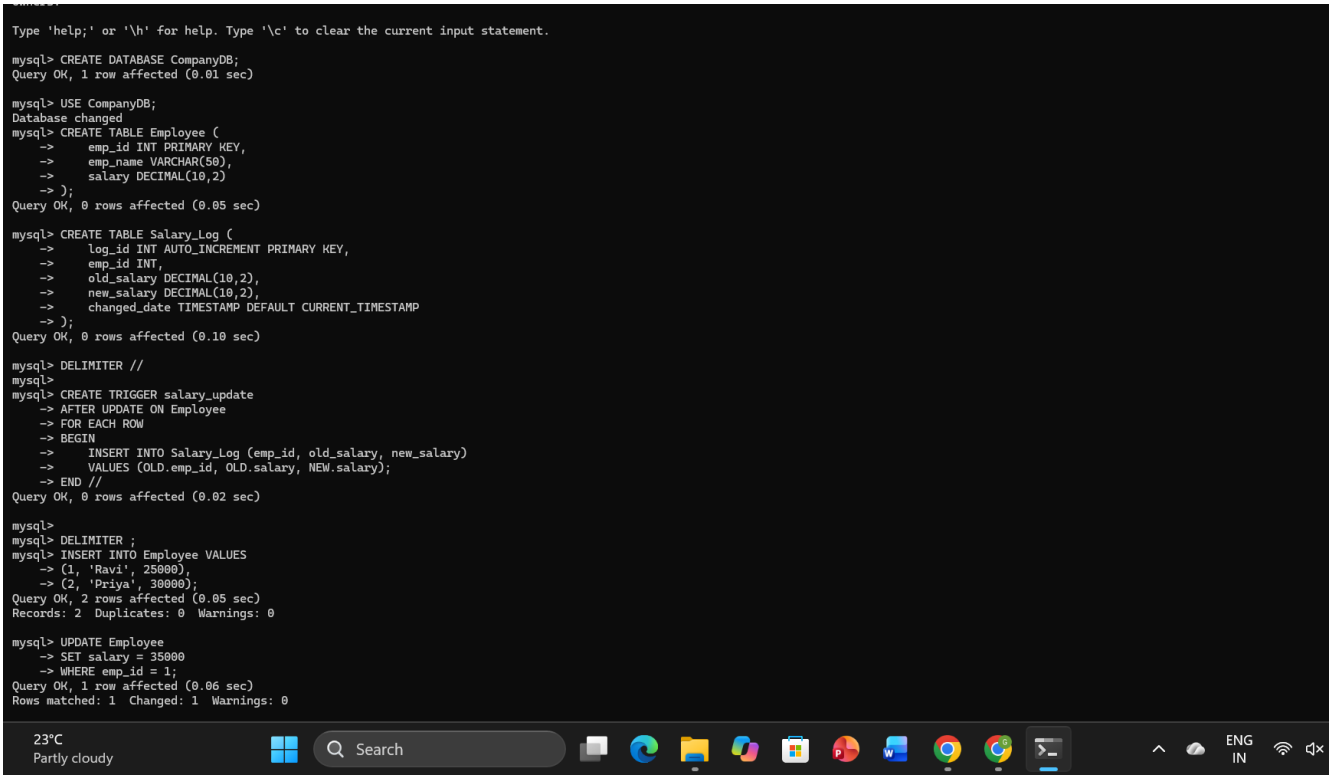
Display employees

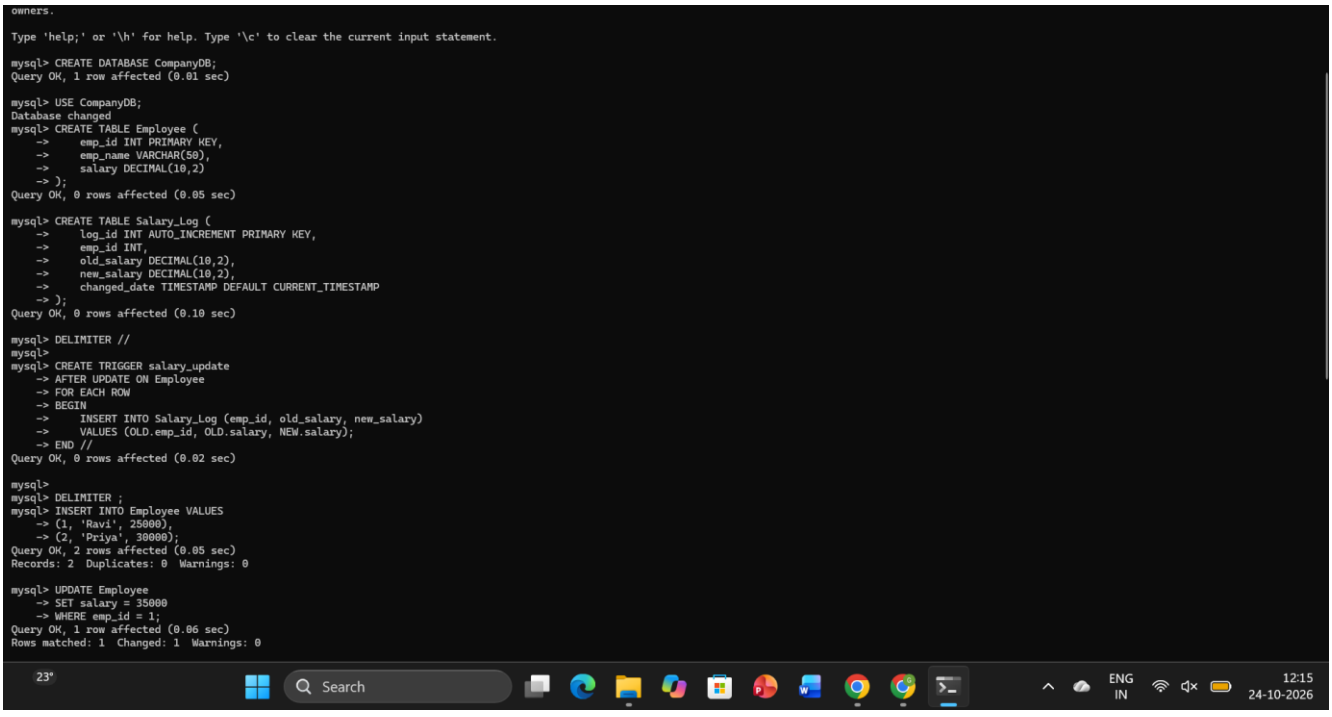
```
SELECT * FROM Employee;
```

Display salary log

```
SELECT * FROM Salary_Log;
```

Expected Output:







Observation

The AFTER UPDATE trigger automatically inserts the old and new salary values into Salary_Log.

Result

Thus, a MySQL trigger was created and executed successfully to maintain a salary change log.